

MetroFlow AI: UI/UX Specification Document

Version: 1.0 (Hackathon MVP Scope) **Date:** August 8, 2026 **Target Domain:** Problem Statement #18 (Smart Infrastructure) **Status:** Approved for System Design

PDF+ 3

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1. Product Overview

Vision

To become the global standard for data-driven municipal traffic management, empowering every city to eliminate preventable gridlock through intelligent, explainable, and accessible signal timing optimization.

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Objectives

- **Automate Bottleneck Detection:** Reduce bottleneck identification time from weeks to under 30 seconds.

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- **Optimize Green Split & Coordination:** Generate offline signal timing plans that yield an estimated 12%-25% reduction in average intersection delay.
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- **Explainable Recommendations:** Provide plain-language narratives powered by AI for every recommended signal change.
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- **Low-Barrier Simulation:** Supply "before vs. after" queue and delay simulation estimates without requiring complex micro-simulation software.
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Users

- **Primary:** Municipal Traffic Engineers, Traffic Operations Center (TOC) Managers.
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- **Secondary:** Smart City Consultants, Transportation Planning Agencies.
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- **Tertiary:** Civil Engineering Contractors, Municipal Operations Directors.
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Scope

MetroFlow AI is strictly an **offline decision-support tool**. It ingests historical traffic volume data, road metadata, and incident logs to automatically detect bottlenecks and compute optimal green-split adjustments.

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Non-Goals:

- The system will NOT directly actuate or interface with live traffic signal controllers.
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- The system will NOT replace heavy microscopic traffic simulation platforms like VISSIM.
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- The system explicitly avoids guaranteed improvement claims, stating all metrics as simulations.
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Core Workflow

1. **Data Ingestion:** Upload historical counts (CSV) and metadata.
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2. **Automated Analytics:** Parse data to identify peak hours and Volume-to-Capacity (V/C) ratios.
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3. **Optimization Engine:** Compute Webster-based cycle lengths and splits.
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4. **Simulated Validation:** Execute M/M/1 queueing models for before/after comparison.
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5. **AI Rationale & Export:** Generate Azure OpenAI engineering narratives and NEMA CSV exports.
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UX Philosophy & Design Principles

- **Engineering Credibility:** Zero "black box" decisions; every numerical output must link to a mathematical formula or explicit AI rationale.
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- **Simplicity:** Intuitive design allowing a trained traffic analyst to complete a workflow in under 5 minutes.
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- **Safety Primacy:** UI must hardcode and prominently display statutory safety bounds (e.g., minimum pedestrian walk times).
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2. Information Architecture

Sitemap & Navigation Hierarchy

- **Global Navigation:** Workspace Selector, Municipal Account Profile, Notifications.
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- **Dashboard (Home):** Active Retiming Projects Grid, System KPIs.
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 - **Project Workspace:**
 - **Step 1: Data Ingestion & Configuration:** File Upload, Metadata Entry, Incident Logs.
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 - **Step 2: Congestion Analytics:** Peak Windows, Bottleneck Heatmap, LOS Charts.
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 - **Step 3: Signal Optimization & Simulation:** Phase Diagram, Slider Controls, Before/After Metrics, Stress Test Slider.
PDF
 - **Step 4: AI Rationale & Export Center:** AI Narrative, Controller CSV Preview, PDF Report Export.
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User Navigation Diagram

Dashboard → Create New Project → Upload CSV & Geometry → Analyze Congestion Map → Optimize Timings → Review Simulation & Safety → Read AI Rationale → Export Reports

3. User Personas

1. Traffic Engineer (Primary)

- **Name:** Marcus Vance, PE
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- **Demographics:** Age 44, M.S. in Civil Engineering, 18 years experience.
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- **Goals:** Clear corridor bottlenecks, lower intersection Level of Service (LOS) delay grades, defend decisions to city council.
PDF
- **Pain Points:** Current static timing plans are 4 years old; Synchro software takes weeks to model; lacks time to manually parse raw CSV logs.
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- **Daily Workflow:** Manually cleans loop-detector data, builds models in legacy software, adjusts splits manually.
PDF
- **Success Metrics:** Reduction in intersection delay time, compliance with MUTCD standards.
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2. Smart City Consultant (Secondary)

- **Name:** Elena Rostova
PDF
- **Demographics:** Age 35, M.S. in Urban Planning & Data Science, 10 years experience.
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- **Goals:** Deliver data-backed signal optimization proposals quickly; demonstrate ROI and emissions reduction.
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- **Pain Points:** Municipal data is fragmented; clients lack budget for expensive real-time adaptive systems (e.g., SCATS).
PDF
- **Daily Workflow:** Aggregates fragmented data into presentation layers for non-technical city leadership.
PDF
- **Success Metrics:** Rapid analysis turnaround, clear visual reporting.
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3. Municipal Authority (Tertiary)

- **Title:** City DOT Leadership
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- **Goals:** Congestion reduction, cost efficiency, public safety compliance, clear reporting metrics.
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- **Pain Points:** Traditional retiming projects cost \$3,000-\$5,000 per intersection and take 6-12 months.
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- **Decision Making Process:** Requires transparent, jargon-free executive summaries and guaranteed safety compliance before approving budgets.
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4. End-to-End User Journey

1. Create Project

- **User Goal:** Establish a new workspace for an arterial corridor.
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- **System Action:** Provisions secure sandbox.
PDF
- **Output:** Empty project dashboard.
- **Next Step:** Upload data.

2. Upload Dataset & Metadata

- **User Goal:** Upload historical 15-minute volume logs and road geometry via CSV.
PDF+ 1
- **System Action:** Parses file format, checks against required fields (Timestamp, Intersection_ID, Volume, Speed_Limit).
PDF
- **Output:** File upload progress bar and validation indicators.
PDF+ 1
- **Next Step:** Validate Data.

3. Validate Data

- **User Goal:** Ensure data is clean and actionable.
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- **System Action:** Detects missing values, duplicates, and unrealistic volumes; interpolates missing 15-minute intervals.
PDF
- **Output:** Warning/Error log (e.g., "Warning: Implausible volume at INT_4. Capped at saturation limit.").
PDF
- **Next Step:** Traffic Analysis.

4. Congestion & Bottleneck Analysis

- **User Goal:** Rapidly identify peak congestion windows and bottlenecks.
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- **System Action:** Applies a rolling 1-hour sum window to find peak boundaries; calculates V/C ratios.
PDF

- **Output:** Corridor heatmap and hourly volume bar charts highlighting V/C > 1.0 (LOS F).
PDF+ 1
- **Next Step:** Signal Optimization.

5. Signal Optimization & Safety Validation

- **User Goal:** Generate better cycle lengths and splits.
PDF
- **System Action:** Runs Webster's formula
 $C0=1-Y1.5L+5$
to compute cycle lengths. Intercepts array with **Safety_Validator** to enforce minimum pedestrian walk times.
PDF+ 3
- **Output:** Recommended Phase Diagram and green split distributions.
PDF
- **Next Step:** Simulation.

6. Simulation & What-If Analysis

- **User Goal:** Verify performance gains and stress-test the plan.
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- **System Action:** Uses M/M/1 queuing logic to calculate delay/queue reductions; responds to +/- 30% volume stress test slider.
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- **Output:** Side-by-side metric cards (Delay, Queue, LOS) and dynamic slider.
PDF+ 1
- **Next Step:** AI Rationale.

7. AI Engineering Explanation & Export

- **User Goal:** Understand the mathematical rationale and export for field use.
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- **System Action:** Passes structured JSON payload to Azure OpenAI to generate plain-text Markdown.
PDF+ 1
- **Output:** Narrative text panel and downloadable NEMA-compliant CSV / PDF summary.
PDF+ 1
- **Next Step:** Field Implementation (External).

5. Screen Specifications

Screen 1: Dashboard & Project Management

- **Purpose:** Overview of all retiming projects and high-level municipal impact.
- **Layout:** Top Header, Left Sidebar, Main Grid area.

- **KPIs:** Total Intersections Analyzed, Average Delay Reduced, Last Plan Update Date.
PDF
- **Components:** Search Bar, Workspace Selector, Action Button ("+ Create New Corridor Retiming Project").
PDF
- **Empty State:** Friendly graphic inviting the user to create their first retiming project.

Screen 2: Data Ingestion & Corridor Configuration

- **Purpose:** Secure upload and data sanitization interface.
- **Layout:** Two-column split layout.
- **Left Panel:** Drag-and-drop box for traffic count CSVs and Incident Logs.
PDF
- **Right Panel:** Interactive node map for defining road geometry (lanes, speed limits, turning ratios).
PDF
- **Validation Rules:** Timestamp format, Volume ≥ 0 , Turn Ratios summing to 1.0.
PDF
- **Error States:** Clear, red textual warnings (e.g., "Error: Row 45 timestamp format invalid") preventing analysis progression.
PDF
- **Primary Action:** "Parse & Analyze Congestion Data" button.
PDF

Screen 3: Congestion Analytics & Bottleneck Dashboard

- **Purpose:** Visual diagnosis of historical traffic failures.
- **Layout:** Top KPI Ribbon, Center Map, Bottom Charts.
- **KPIs:** Peak Hour Identified, Worst Intersection V/C Ratio, Corridor Average Delay.
PDF
- **Visualizations:** Arterial corridor map (color-coded red/orange for bottlenecks), Hourly volume bar charts with shaded peak overlay boxes.
PDF+ 1
- **Expected Output:** Intersection #4 flagged as primary bottleneck with V/C > 1.0 (LOS F).
PDF+ 1

Screen 4: Optimization & Simulation Workbench

- **Purpose:** The core engineering interface for testing recommended timings.
- **Layout:** Split View.
- **Left Side (Parameters):** Interactive phase diagram, slider controls for Cycle Length (90s - 150s) and Phase Green Splits.
PDF
 - *OPTIONAL HACKATHON ENHANCEMENT:* Interactive Time-Space progression diagram.
PDF+ 1

- **Right Side (Results):** Side-by-side "Before vs. After" comparative metrics (Control Delay, Queue Length, Intersection LOS).
PDF
- **Interactive Element:** "What-if" Volume Surge Stress Test slider (+/- 30%).
PDF+ 1
- **Tooltips:** Explicit disclaimers on queuing models stating "Simulated Estimate - Assumes Stable Flow".
PDF
- **Safety Alerts:** Live warnings if manual adjustment breaches minimum pedestrian walk times.
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Screen 5: AI Rationale & Report Export Center

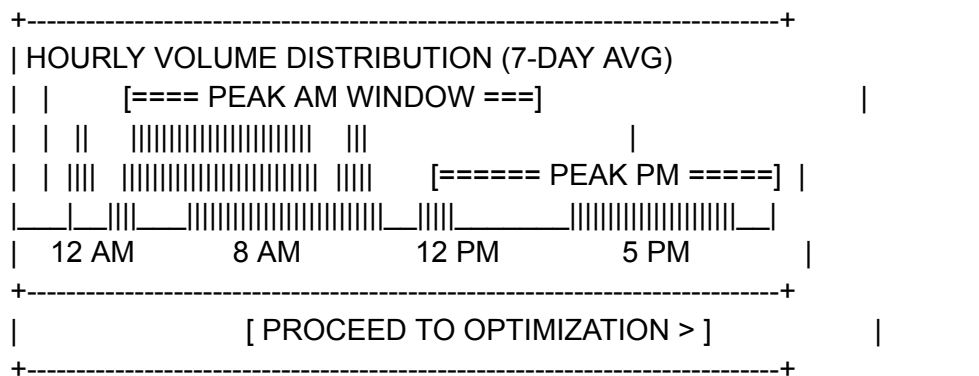
- **Purpose:** Final review, explanation, and extraction of results.
- **Layout:** Two-column view.
- **Left Panel:** Markdown-formatted plain-language narrative explaining adjustments (powered by Azure OpenAI).
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- **Right Panel:** Live preview of the Controller Parameter Table (NEMA format) and PDF document.
PDF
- **Buttons:** "Download Controller Timing Sheet (CSV)", "Download Executive PDF Report".
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- **Expected Outputs:** A 2-3 paragraph plain language summary citing explicit simulation assumptions.
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6. Wireframes

Screen 3: Congestion Analytics & Bottleneck Dashboard

Plaintext

+-----+																		
[MetroFlow AI Logo]		Workspace: Main St Corridor				[User Profile]												
+-----+																		
[KPI 1: Peak Hour Identified: 07:45 AM - 09:15 AM]																		
[KPI 2: Worst Intersection V/C Ratio: 1.12 (INT_04)]																		
[KPI 3: Corridor Avg Delay: 68.4 s/veh (LOS E)]																		
+-----+																		
CORRIDOR HEATMAP (GIS VIEW)																		
(INT_1)=====(INT_2)=====(INT_3)=====[INT_4]=====(INT_5)																		
[LOS B]		[LOS C]		[LOS C]		[LOS F]		[LOS D]										

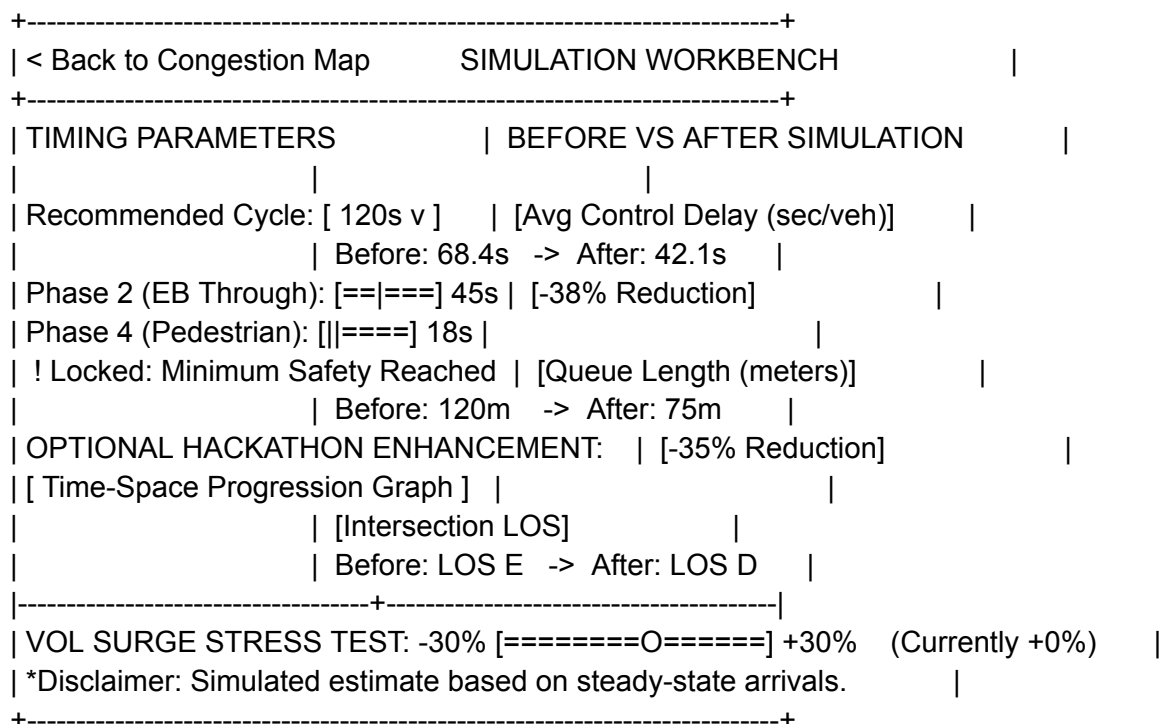


Explanation: Top ribbon highlights the most critical findings. The heatmap visually grounds the engineer in the spatial reality of the corridor. The bottom chart provides the temporal evidence for the peak window detection, which feeds into the optimization engine.

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Screen 4: Optimization & Simulation Workbench

Plaintext



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Explanation: The split layout ensures that any adjustment made on the left side (Parameters) instantly reflects in the mathematical output on the right side (Simulation). The safety guardrail is visually enforced with a lock icon. The stress test slider provides interactive validation.

PDF+ 3

7. Dashboard Design

The dashboard must reflect premium enterprise SaaS analytics, avoiding cluttered visuals.

- **KPIs:** Placed at the top for immediate executive visibility. Use large typography for numbers (e.g., **1.12 V/C**) and smaller semantic text for units.
- **Cards:** Use subtle shadows and rounded corners (Border radius: 8px) to contain discrete metrics (e.g., Delay Reduction).
- **Heatmaps/Maps:** Use a dark-mode base map by default to allow brightly colored intersection nodes (Red = Bottleneck, Green = Free Flow) to stand out sharply.
PDF
- **Alerts/Status Indicators:** Strict semantic coloring. Red for LOS F/Safety breaches, Yellow for LOS D-E, Green for LOS A-C.
- **Recommended Actions:** Action buttons should be high-contrast and logically placed at the bottom right of the screen flow to guide the user to the next step.
- *Why every widget exists:* To map directly to a user story, prioritizing rapid decision-making and defensible engineering reporting over decorative elements.
PDF

8. Data Visualization Guide

- **Traffic Volume & Hourly Trend: Bar Chart with Overlay Box.** The bar chart accurately reflects discrete hourly volume aggregation, while a shaded overlay clearly delineates the algorithmic clustering of the "Peak Window".
PDF+ 1
- **Queue Length & Delay: Side-by-Side Metric Delta Cards.** The core value proposition is the "Before vs. After" comparison. A simple side-by-side layout with a percentage reduction badge is the fastest way to communicate ROI.
PDF
- **Congestion Severity (LOS) & V/C Ratio: Node-based GIS Heatmap.** Traffic engineers think spatially. Mapping V/C ratios directly onto intersection nodes on a linear corridor instantly reveals where the system is failing (e.g., INT_04).
PDF+ 1
- **Signal Timing Comparison: Horizontal Stacked Bar Chart (Phase Diagram).** Maps directly to industry-standard NEMA ring-and-barrier diagrams, showing how the total cycle length is sliced into green, yellow, and red intervals.
- **OPTIONAL HACKATHON ENHANCEMENT - Simulation / Progression: Time-Space Diagram.** Shows distance on the Y-axis and time on the X-axis, allowing engineers to visually trace the green wave (platoon progression) across the arterial.
PDF+ 1

9. GIS / Map Experience

- **Map Style:** Minimalist vector map (e.g., Mapbox Dark) stripped of unnecessary POIs to focus solely on road geometry.
- **Intersection Markers:** Circular nodes located at exact geocoordinates.

- **Severity Colors:**
 - V/C < 0.85 (Green)
 - V/C 0.85 - 0.99 (Yellow)
 - V/C > 1.0 (Red)
- **Hover Cards:** Hovering over a node displays its ID (e.g., INT_04), Approach Direction (Eastbound), Lanes, and Baseline Control Delay.
- **Legend:** Standardized HCM LOS color coding (A-F).
- **Map Interactions:** Pan, Zoom, and Click-to-select intersection for deep-dive analysis.

10. Design System

- **Typography:** Inter or Roboto. Sans-serif, highly legible at small sizes for data tables. Monospace font (e.g., Fira Code) for raw JSON or CSV previews.
- **Primary Colors:** Professional Blue (`#0F52BA`) for primary actions and highlights to establish engineering credibility.
- **Semantic Colors:**
 - Success/Safe/LOS A: Green (`#10B981`)
 - Warning/LOS D: Yellow/Orange (`#F59E0B`)
 - Critical/LOS F/Safety Breach: Red (`#EF4444`)
- **Spacing & Grid:** 8pt grid system. 24px padding within cards.
- **Components:**
 - **Buttons:** Solid fill for primary, outlined for secondary.
 - **Forms:** Floating label text inputs for geometry entry.
 - **Tables:** Striped rows for CSV data validation previews to aid horizontal reading.
- **Border Radius:** 8px for modern, soft-edged professional UI.
- **Shadows:** Soft drop-shadows on floating elements (hover cards, tooltips).

11. Accessibility

- **WCAG 2.1 AA Compliance:** Minimum contrast ratio of 4.5:1 for all text against backgrounds.
- **Color Contrast:** Critical UI elements (like bottlenecks) must not rely on color alone; use icons or text labels (e.g., explicitly stating "LOS F") alongside the red color.
- **Keyboard Navigation:** Fully tab-navigable interface, prioritizing the sequential user journey flow.
- **Focus States:** High-visibility blue focus rings (`box-shadow: 0 0 0 3px rgba(15, 82, 186, 0.5)`) around active sliders and inputs.
- **Dark Mode:** Supported natively, as TOC command centers often operate in low-light environments.

12. UX Details

- **Empty States:** When a project is created but no CSV is uploaded, display an illustration of a traffic intersection with a clear "Upload CSV to Begin" prompt.
- **Loading States:** Use skeleton screens matching the final layout (e.g., pulsing grey bars where the bar chart will appear) instead of generic spinners during the 5-second optimization calculation.
PDF
- **Error Handling:** Inline validation during CSV upload. If turning ratios sum to > 1.0 , highlight the specific cell in red and disable the "Analyze" button.
PDF
- **Toast Notifications:** Non-intrusive bottom-right toasts for background actions (e.g., "NEMA CSV Exported Successfully").

13. Microinteractions

- **File Upload:** Dropzone border dashes animate and background color dims slightly upon `dragenter`.
- **Charts:** Bars in the hourly volume chart grow from the X-axis upon page load.
- **Simulation Slider:** Moving the "What-If" volume surge slider causes the queue length numbers to smoothly tick up/down rather than flashing instantly, visually reinforcing the mathematical relationship.
PDF+ 1
- **Safety Lock:** Attempting to drag a green split slider below the pedestrian minimum triggers a subtle horizontal "shake" animation on the slider handle, accompanied by a visual lock icon.
PDF

14. Developer Handoff

Step 3: Congestion Analysis API

- **Expected Input Data:** Aggregated 15-minute volume JSON arrays.
PDF
- **Expected Backend API:** Python temporal cluster analysis.
PDF
- **Output Data:** `peak_window_start`, `peak_window_end`, `max_vc_ratio`, `bottleneck_node_id`.
PDF

Step 4: Optimization API

- **Expected Input Data:** Critical lane volumes, clearance lost times (L), min pedestrian times.
PDF

- **Expected Backend API:** Deterministic Python engine (Webster's formula).
PDF
- **Output Data:** JSON array of phase allocations (`phase_2_green`: 45s).
- **Validation Rules:** `Calculated_Green[Phase] >= Min_Ped_Walk[Phase]`.
PDF

Step 5: AI Rationale API

- **Expected Input Data:** Structured JSON containing before/after metrics.
- **Expected Backend API:** Azure OpenAI prompt endpoint.
PDF+ 1
- **Expected Result:** Markdown string (e.g., "Green time for Phase 2 was increased by 14 seconds... Minimum pedestrian clearance times were strictly preserved at 18 seconds."). **Constraint:** LLM must NOT calculate math.
PDF+ 3

15. Requirement Traceability Matrix

Requirement	Screen	UI Component	User Action	Expected Result
FR-A01: CSV Upload PDF	Data Ingestion	Drag/Drop Zone	Upload File	Parses file, validates format. PDF
FR-B02: Peak Congestion PDF	Congestion Map	Bar Chart Overlay	View Screen	Highlights AM/PM peaks. PDF+ 1
FR-B01: V/C Ratio > 1.0 PDF	Congestion Map	Heatmap Node	Hover over INT_04	Displays V/C 1.12, flags Bottleneck. PDF+ 1
FR-C04: Safety Lower-bounds PDF	Optimization	Slider / Lock Icon	Try reducing green below min	Slider stops, displays "Locked" warning. PDF+ 1
FR-D01: Queuing Simulation PDF	Optimization	Metric Delta Cards	View Screen	Shows before/after delay reductions. PDF

FR-D04: What-if Volume Surge	Optimization	Interactive Slider	Drag slider to +20%	UI updates dynamically, recalculates queue.
PDF				PDF+ 1
FR-E01: AI Explanations	AI Rationale	Markdown Panel	Read Text	Plain English explanation of splits.
PDF				PDF+ 1
FR-E03/04: Report Exports	AI Rationale	Download Buttons	Click "Export PDF"	Downloads executive summary and NEMA CSV.
PDF				PDF

16. Hackathon Demo Flow

Ideal 5-Minute Script:

- Transition 1 (Screen 1 & 2 - 45 seconds):**
 - Action:* Presenter logs in and drags a CSV onto the upload zone.
 - Narration:* "Cities waste months manually identifying traffic patterns. We just uploaded 7 days of raw count data for Main Street. MetroFlow AI instantly cleans and processes it."
- Transition 2 (Screen 3 - 60 seconds):**
 - Action:* Screen reveals the Corridor Heatmap. Presenter hovers over Intersection 4.
 - Narration:* "Instantly, we see our AM Peak is 7:45 to 9:15, and our primary bottleneck is Intersection 4, failing with a V/C ratio of 1.12. Let's fix this."
- Transition 3 (Screen 4 - 90 seconds):**
 - Action:* Click to Optimization Workbench. Adjust the What-If slider. Attempt to reduce pedestrian time.
 - Narration:* "Our offline engine applies Webster's method, redistributing green time to the saturated Eastbound approach. Watch as I drag the 'What-If' slider to simulate a 20% traffic surge—the model recalculates dynamically. If I try to reduce green time below the pedestrian minimum walk time... the system locks it. Safety is hardcoded."
- Transition 4 (Screen 5 - 60 seconds):**
 - Action:* Reveal the AI summary text and click the Export PDF button.
 - Narration:* "Finally, we use Azure OpenAI to translate these complex queuing dynamics into a plain-language executive summary, explicit assumptions included. We export the NEMA controller file, and we're ready for field programming in under 5 minutes."

5. **Judge Takeaway:** The solution is highly practical, directly solves municipal pain points, enforces safety, and is ready to deploy offline without hardware risk.

17. Judge Review

Acting as a Hackathon Judge evaluating this UI/UX Specification:

Weaknesses & Limitations

- **Reliance on Static CSVs:** The UI relies on the user uploading clean, structured CSV data, whereas real municipal loop detector data is notoriously messy.
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- **Queuing Model Limitations:** UI must make it incredibly obvious that M/M/1 queuing equations under-predict delay during extreme oversaturated conditions ($V/C > 1.2$).
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Missing Features / UI Considerations

- **Pedestrian & Bicycle Phasing:** The UI heavily emphasizes vehicular throughput, but engineers legally require detailed interfaces for Flashing Don't Walk (FDW) and bicycle minimums.
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Potential Judge Questions

- *Question:* "How do you ensure the AI doesn't hallucinate a dangerous signal timing?"
- *Answer:* The UI and architecture clearly separate math from text. Python deterministic engines handle all mathematical generation and safety locks; Azure OpenAI solely reads a structured JSON to generate a narrative summary.
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Optional Enhancements Evaluated

- **Automated CSV Data Cleaning UI:** Highly recommended to add a pre-analysis screen where users can visualize missing timestamps being interpolated.
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- **Time-Space Progression Diagram:** The inclusion of this diagram in the UI greatly elevates the engineering credibility of the platform.
PDF+ 1

Hackathon UI/UX Scorecard

- **Innovation:** 8/10 (Smart use of LLM for explanation rather than generation)
- **Usability:** 9/10 (Clear linear flow, intuitive split-pane optimization screen)
- **Engineering Credibility:** 10/10 (Prominent safety locks and explicitly cited assumptions)
- **Demo Quality:** 9/10 (Fast time-to-insight, highly visual heatmap)

- **Professional Appearance:** 9/10 (SaaS-tier dashboard aesthetics)
- **Hackathon Readiness:** 10/10 (Scoped strictly to MVP guidelines)